StockEvent Lens — Batch 012 Intelligence Dashboard

Articles 056–060 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_12 (Articles 056-060)`

TEST 056
Original — 100% Base
ASTS falls 6% to $59.24 amid space selloff; joint US-Japan backing for Rakuten deal[cite: 87, 89].

TEST 057
Repeat — 96% Overlap
Re-syndicated wire report on ASTS 6% drop and moving average technicals[cite: 87, 90].

TEST 058
Update — +22% New Info
ASTS rebounds 2% to $60.45 as Rakuten confirms H1 2027 commercial service launch[cite: 87, 91].

TEST 059
Conflict — Dispute (+4% / US)
Contradicts price and technicals: ASTS rose 4% to $64.80 with US-only backing & reversed death cross[cite: 87, 92].

TEST 060
Different — 0% Overlap
Unrelated S&P pullback analysis with subdued VIX below 16 starting Group 024[cite: 87, 88].

Technical Analysis: Dense Vector Blindness vs. Directional & Government Backing Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *directional stock sentiment or geopolitical alignment fidelity*[cite: 87]. Changing price trajectories and backing scope (e.g., -6% drop to $59.24 with joint US-Japan backing vs +4% surge to $64.80 with US-only backing in test_059) yields high vector similarity (~0.89) because both texts share identical surrounding context tokens ("AST SpaceMobile", "Rakuten Group", "SpaceX", "death cross", "moving average")[cite: 87, 89, 92].

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard directional price move or geopolitical attribution dispute[cite: 87].
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "fell 6% to $59.24 with joint US-Japan backing" and "rose 4% to $64.80 with US-only backing" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this directional conflict[cite: 87].

2. LLM Evaluation Override
To prevent missing critical price movement disputes, the LLM reasoning layer performs explicit directional and entity attribute-value matching, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected[cite: 87].

Article Relationship & Percentage Breakdown (Batch 012)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_056_original`	Original (%5=1)	ASTS, down 6% to $59.24, joint US-Japan backing, Rakuten, death cross[cite: 87, 89]	100%	Baseline	Establishes breaking story frame on ASTS stock slide and joint government backing[cite: 87, 89].
`test_057_repeat`	Repeat (%5=2)	ASTS fell 6% to $59.24, 20-day SMA $60.15, RSI 48.08, July death cross[cite: 87, 90]	96%	0% New Facts	Rephrased wire report detailing identical price technicals and government backing[cite: 87, 90].
`test_058_update`	Update (%5=3)	**Rebounds 2% to $60.45**, Thursday recovery, Rakuten H1 2027 commercial service launch[cite: 87, 91]	83%	+22% Operational Facts	Advances event state with Thursday price recovery and Rakuten's H1 2027 commercial launch confirmation[cite: 87, 91].
`test_059_conflict`	Conflict (%5=4)	**Rose 4% to $64.80 dispute**, US-only backing, RSI 62.4, death cross reversed[cite: 87, 92]	88% Match	$\Delta F_{contradiction}$ = 1.0	Directly disputes price direction and technicals, asserting ASTS rose 4% to $64.80 with US-only backing[cite: 87, 92].
`test_060_different`	Different (%5=0)	S&P 500 pullback, 10-year Treasury yield >5.3%, VIX below 16, SKEW 141[cite: 87, 88]	0%	Group 024 Trigger	Zero domain overlap; triggers standalone Group 024 event cluster[cite: 87, 88].

Group 023 — Consolidated Event Summary

4 Articles (056–059)

AST SpaceMobile experienced stock trading activity tied to government backing and partner updates for its Rakuten Group satellite venture (**Articles 056–057**)[cite: 87, 89, 90]. Operational momentum progressed as **Rakuten confirmed commercial service launch for H1 2027, lifting ASTS stock 2% to $60.45** (**Article 058**)[cite: 87, 91]. However, stock trajectory and government backing details are in direct conflict: original reports noted ASTS fell 6% to $59.24 amid a sector selloff with joint US-Japan backing, whereas a contradictory report claims **ASTS rose 4% to $64.80 with US-only backing, reversing its death cross** (**Article 059** ; $\Delta F_{contradiction}$ = 1.0)[cite: 87, 92].

✓ Update Preserved: Rakuten Confirmed H1 2027 Service Launch
✓ Conflict Preserved: Down 6% ($59.24) vs Up 4% ($64.80)
✓ Duplicate Removed: Article 057 Consolidated

Group 024 — Standalone Summary

1 Article (060)

Wall Street pulled back as 10-year Treasury yields topped 5.3% and oil passed $100 , yet the VIX remained subdued below 16 while SKEW reached 141, signaling a valuation reset rather than broad market panic for tech leaders like NVIDIA (**Article 060**)[cite: 87, 88].

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 023

Batch 012 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

test_056_original.json — Group 023

```
{
  "id": 62228481,
  "headline": "Why Is AST SpaceMobile Stock Falling on
  "symbols": ["ASTS", "NVDA", "SPCX"],
  "summary": "Why is ASTS stock falling Wednesday? AST
}
```

Establishes -6% drop & joint US-Japan backing[cite: 87, 89].

test_058_update.json — Group 023

```
{
  "id": 90000005,
  "headline": "AST SpaceMobile Rebounds Thursday After
  "symbols": ["ASTS", "SPCX"],
  "summary": "AST SpaceMobile rose about 2% to $60.45 a
}
```

Adds Rakuten H1 2027 commercial launch confirmation[cite: 87, 91].

test_059_conflict.json — Group 023

```
{
  "id": 90000006,
  "headline": "Why Is AST SpaceMobile Stock Rising on t
  "symbols": ["ASTS", "NVDA", "SPCX"],
  "summary": "AST SpaceMobile rose about 4% to $64.80,
}
```

Disputes -6% vs +4% price move and US-only backing[cite: 87, 92].